REQUIEREMENT SPECIFICATION

Electrodes and Distribution

- The electrodes selected during our testing must be used, ensuring that the chosen material is optimal for the defined use case.

- Uniform distribution of electrodes across the torso:

  - **30 electrodes** strategically placed to capture signals in the required positions.

  - **4 references** positioned to maintain constant contact with the patient's torso.

Signal Conduction

- Cables must adequately conduct signals from the electrodes to the connector without compromising signal quality or distribution.

- Cable material: resistant to clinical use, flexible, and compatible with the characteristics of the captured signals.

Connector and Electronics

- The design must include a **single connector** that consolidates all cable connections and enables direct integration with the external electronics.

- The connector must be robust, reliable, and easy to connect and disconnect.

Support for Electronics

- Include a **secure and accessible pocket** to house the electronics, with dimensions suitable for our device.

- The pocket should be designed to facilitate the use and maintenance of the vest.

Regulatory Compliance

- The vest must comply with applicable safety and regulatory standards for medical devices, including:
    - **Electrical safety**
    - **Electromagnetic compatibility (EMC)**
    - **Chemical characterisation**
    - **Biocompatibility**

Design and Sizing

- Provide an ergonomic design that is comfortable for patients during clinical use.
- Initial prototype in a **standard size**, with provisions for larger sizes in future production phases.

Maintenance and Reuse

- The vest must be **washable and reusable in a clinical setting**, after the electronics are removed.
- Materials should be durable and resistant to washing cycles according to hospital protocols.